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Unit – 4 : Session – 11 : SLO – 1 | B-Trees — Insertion Algorithm

■ I. Inserting a New Element in a B Tree

- In a B tree, all insertions are done at the LEAF NODES.
- Step 1: Search the B tree to find the LEAF NODE where the new key value should be inserted.
- Step 2: If the leaf node is not full, that is, it contains less than $m-1$ values, then insert the new element in the node keeping the node's elements ordered.
- Step 3: If the leaf node is FULL, that is, the leaf node already contains $m-1$ key values, then:
 - a) Insert the new value IN ORDER into the existing set of keys,
 - b) SPLIT the node at its median into two nodes (note that the split nodes are half full), and
 - c) Push the median element up to its PARENT NODE. If the parent's node is already full, then
- SPLIT the parent node by following the same steps.